

Amortizing Physics-Informed PDE Surrogates over Initial and Boundary Conditions, Not Just Coefficients

Anonymous submission

Abstract

Parametric physics-informed neural networks (PINNs) solve a family of parameterized PDEs with one network by encoding the PDE coefficients into a latent code, amortizing training across the coefficient space. But they condition on the coefficients *only*: the initial condition (IC) and boundary condition (BC) enter through the loss and are fixed once trained, so a trained model cannot be queried at a new IC or BC—exactly the situation that arises in design, control, and uncertainty quantification. We recast the coefficient encoder as a *problem-specification encoder* that also ingests the IC and BC, and study what makes such conditioning generalize. A single residual-trained model conditioned on the full specification generalizes to unseen IC parameters, an unseen IC *functional family*, unseen coefficient/IC combinations, two BC types, and multiple PDE families, reducing relative L_2 error on held-out ICs by roughly $2\text{--}9\times$ (axis-dependent) versus the coefficient-only surrogate across five 1D convection–diffusion–reaction families, at matched budget and with no meaningful regression on the original coefficient-only task. We then ask *how* to represent the IC, comparing a harmonic (Fourier) basis encoding against an information-matched point-sample encoding. Their difference is governed by the *transport* content of the dynamics: the harmonic encoding clearly improves cross-family generalization for transport-dominated problems (convection: relative L_2 0.503 vs. 0.614) and its margin over point samples grows *monotonically* with the convection coefficient, whereas for reaction-dominated (pointwise) dynamics dense point samples are at least as good—a distinction we explain mechanistically (transport is a spectral phase shift, trivial in the harmonic basis but aliased under sampling; reaction is pointwise and representation-neutral) and use to predict which representation is preferable. Conditioning-removed ablations, a latent-space analysis, an out-of-distribution study mapping a finite-band failure, a boundary-condition extension with an authored and validated non-periodic reference, and joint support of three PDE families complete the picture. We thus reframe conditioning in parametric neural-PDE surrogates as a question of the full specification *and* its representation, and characterize honestly when a structured (harmonic) encoding helps.

1 Introduction

Physics-informed neural networks (PINNs) [Raissi et al., 2019] solve a PDE by training a network $u_\Theta(x, t)$ to minimize the PDE residual at collocation points together with initial- and boundary-condition penalties. Their appeal is that they need no precomputed solution data; their cost is that a fresh network must be trained for every new problem. This is prohibitive in the multi-query settings that motivate surrogate modeling—design optimization, control, and uncertainty quantification—where a PDE must be solved at very many points of a *problem-specification space*.

A productive response has been to *amortize* across that space with a single conditioned network. The parameterized-PINN approach of Cho et al. [2024a] is representative and is our starting point: rather than appending the PDE coefficients μ to the input coordinates, it maps μ through a dedicated encoder to a latent code $\mathbf{h}_p = g_{\theta_p}(\mu)$, concatenates it with a coordinate embedding $\mathbf{h}_c = g_{\theta_c}(x, t)$, and decodes the solution, $u_\Theta(x, t; \mu) = g_{\theta_g}([\mathbf{h}_c; \mathbf{h}_p])$. Encoding the coefficients into a latent manifold, rather than treating them as extra coordinates, markedly improves accuracy and overcomes PINN failure modes at large convection/reaction [Krishnapriyan et al., 2021], and a single model covers an entire coefficient range.

The gap. Such surrogates condition on *one* axis of the specification. The problem specification of a parameterized PDE is a triple—coefficients, initial condition (IC), and boundary condition (BC)—and in the applications above the IC and BC vary as much as the coefficients: sources and starting profiles change, and the boundary regime changes with the geometry or the physics. In the coefficient-conditioned surrogate the IC enters only through the data-matching loss term and the BC only through the boundary loss term; both are *fixed* for a trained model. Nothing about the IC or BC is an input to the encoder, so the model cannot be evaluated at a new IC or BC without retraining. This is a representational limitation, not a tuning issue: the surrogate has amortized one axis and baked in the other two.

This paper. We turn the coefficient encoder into a *problem-specification encoder* $g_{\theta_p}(\mu, \xi_{\text{IC}}, \xi_{\text{BC}})$ that ingests representations of the IC and BC alongside the coefficients, while holding fixed the decoder, the coordinate encoder, the concatenation injection, the residual-plus-condition training objective, and the training budget. Two questions follow. First, *does* conditioning on the full specification let one model generalize across ICs and BCs? We find that it does, and strongly: a single model reduces held-out-IC error several-fold relative to the coefficient-only surrogate on every generalization axis of five PDE families. Second, *how* should the specification be represented? Here we report a nuanced, mechanism-driven answer. Comparing a harmonic (Fourier) basis encoding of the IC against an information-matched point-sample encoding, we find their difference is governed by the *transport* content of the dynamics. Transport is a spectral phase shift—diagonal and trivial in the harmonic basis, but aliased and hard to undo from point samples—so the harmonic encoding clearly improves cross-family generalization for transport-dominated problems, and its margin grows monotonically with the convection coefficient. Reaction is pointwise and representation-neutral, so for reaction-dominated dynamics dense point samples are at least as good. We make this precise and use it to predict which representation to prefer.

Contributions.

- A problem-specification encoder that conditions a residual-trained PINN surrogate jointly on the PDE coefficients, the initial condition, and the boundary condition, keeping the decoder, coordinate encoder, injection, loss, and budget fixed, so that a single model becomes *queryable* across the whole specification. It generalizes to unseen IC parameters, an unseen IC functional family, unseen coefficient/IC combinations, two boundary-condition types, and multiple PDE families, cutting held-out-IC error by roughly $2\text{--}9\times$ (axis-dependent) relative to the coefficient-only surrogate at matched budget with no meaningful regression on the original task.
- A controlled representation study at matched information and capacity—harmonic basis coefficients vs. point samples of the same IC—establishing that the harmonic encoding’s benefit is governed by transport: a clear cross-family win on convection and a margin that increases monotonically with the convection coefficient, and an honest null result on reaction-dominated families where point samples are equally good. Conditioning-removed ablations and a non-degeneracy check confirm that both encoders genuinely use the IC and that the harmonic gain, where present, is representational.
- A mechanism analysis (a latent-space diagnostic and an error-versus-novelty analysis) that connects the transport picture to generalization and *predicts* the failure regime—sharp, high-frequency ICs beyond the resolved harmonic band—which we map and show persists under a $2\text{--}4\times$ increase of capacity, epochs, and collocation against a convergence-checked reference.
- Extensions: a single model that conditions on and generalizes across two boundary-condition types (with an authored, validated non-periodic reference solver) and jointly supports three PDE families with transfer to unseen coefficient/IC/BC compositions, plus an honest account of what does not transfer.

2 Related work

Parametric PINN surrogates. A line of work conditions a single PINN on PDE parameters to avoid per-instance retraining: hypernetworks that emit PINN weights from the parameters [de Avila Belbute-Peres et al., 2021, Cho et al., 2024b], meta-learned initializations [Finn et al., 2017], and the latent-code factorization of Cho et al. [2024a] that we build on. All of these amortize over *coefficients*; the initial and boundary conditions remain properties of the loss, fixed per trained model. Our contribution is orthogonal to the amortization mechanism: we change *what the encoder conditions on* and, crucially, *how* the initial/boundary conditions are represented, so that one model is queryable across coefficients, ICs, and BCs.

Operator learning. Neural operators learn a mapping between function spaces—typically from the input function (IC, forcing, or coefficient field) to the solution—from a dataset of solved instances [Lu et al., 2021, Li et al., 2021, Kovachki et al., 2023, Seidman et al., 2022]. DeepONet’s branch network [Lu et al., 2021] ingests the input function through its sampled values; the Fourier neural operator [Li et al., 2021] parameterizes the operator with spectral convolutions. Physics-informed DeepONets [Wang et al., 2021, 2022] train the operator on the PDE residual rather than on solution data, and thereby also condition on the IC/forcing. Two differences frame our work. First, the standard branch-net representation of the input function is precisely a *point-sample* encoding; we show that for cross-family IC generalization a *harmonic* encoding of the same function is markedly better, and we isolate this effect at matched information and capacity. Second, we retain the coefficient-encoder factorization of a parametric PINN and condition *jointly* on coefficients, IC, BC, and PDE type, with an analysis of *why* the structured conditioning generalizes rather than a solution-error table alone.

Conditional neural fields. Representing a datum as a coordinate network modulated by a latent code is central to implicit neural representations [Park et al., 2019, Sitzmann et al., 2020, Mildenhall et al., 2020, Dupont et al., 2022]. A parametric PINN is a physics-informed conditional neural field: the specification code plays the role of the shape/instance code. We adopt this view and ask which *encoding* of the conditioning function produces a code that generalizes; harmonic features are a natural, physics-aligned answer for periodic transport–diffusion dynamics, complementary to random Fourier *coordinate* features [Tancik et al., 2020], which address spectral bias in (x, t) rather than the representation of the specification.

Generalization across environments and PDEs. Context-adaptation methods learn a shared model with per-environment codes that generalize to new dynamical regimes [Yin et al., 2021, Kirchmeyer et al., 2022], and message-passing/graph solvers condition on discretized fields [Brandstetter et al., 2022]. These are largely data-driven; we operate in the residual-trained, data-free PINN setting and analyze the harmonic structure of the learned specification manifold. Benchmarks such as PDEBench [Takamoto et al., 2022] motivate broad specification spaces. Our conditioning combined with the coefficient encoder is, to our knowledge, the first to demonstrate—and mechanistically explain—joint generalization of a single residual-trained surrogate across coefficients, IC parameters, an unseen IC family, boundary-condition type, and PDE family.

3 A problem-specification encoder

3.1 Setup and the factorization we retain

We consider the one-dimensional convection–diffusion–reaction (CDR) family on the periodic domain $x \in [0, 2\pi)$, $t \in [0, 1]$,

$$u_t + \beta u_x - \nu u_{xx} - \rho u(1 - u) = 0, \quad u(x, 0) = u_0(x), \quad (1)$$

with coefficients $\boldsymbol{\mu} = (\beta, \nu, \rho)$ and the Fisher reaction $\rho u(1 - u)$ [Krishnapriyan et al., 2021]. Its specializations—reaction ($\beta=\nu=0$), reaction–diffusion ($\beta=0$), and convection ($\nu=\rho=0$)—have exact references (closed-form logistic; operator splitting [Strang, 1968]; exact spectral evolution), which we reuse unchanged (§4). A *problem specification* is the triple $s = (\boldsymbol{\mu}, u_0, \mathcal{B})$ of coefficients, initial condition, and boundary condition $\mathcal{B}$.

We retain the parametric-PINN factorization of Cho et al. [2024a] exactly:

$$u_{\Theta}(x, t; s) = g_{\theta_g}([\mathbf{h}_c; \mathbf{h}_p]), \quad \mathbf{h}_c = g_{\theta_c}(x, t), \quad \mathbf{h}_p = g_{\theta_p}(\cdot), \quad (2)$$

with the coordinate encoder g_{θ_c} ($2 \rightarrow 50 \rightarrow 50 \rightarrow 50$), the residual decoder g_{θ_g} (six residual blocks of width 50 with a hyper-shift term), and the concatenation injection all held fixed. **The only component we change is the input to g_{θ_p} .**

3.2 From a coefficient encoder to a specification encoder

The reference encoder reads the three coefficients, $\mathbf{h}_p = g_{\theta_p}(\boldsymbol{\mu})$. We widen its input to

$$\mathbf{h}_p = g_{\theta_p}([\boldsymbol{\mu}; \xi_{\text{IC}}(u_0); \xi_{\text{BC}}(\mathcal{B})]), \quad (3)$$

keeping the encoder’s depth, hidden widths (150), output width (50), and activations identical; only the first layer grows to accept the specification features. All features travel on a floating-point path (we deliberately avoid the value-truncating integer cast used for coefficients in the reference data pipeline, which would destroy a continuous amplitude, width, or frequency). The decoder’s hyper-shift term reads only $[\boldsymbol{\mu}; x; t]$, never the IC/BC features, so the decoder is structurally unchanged and identical across all encoders we compare; the specification information reaches the decoder solely through $\mathbf{h}_p$.

Initial-condition representation. This is the crux. Given u_0 sampled on the reference 256-point grid, we compare two encodings of *the same function*:

- **Harmonic (ours).** The low-order real Fourier coefficients $\xi_{\text{IC}}^{\text{harm}} = (\hat{a}_0, \{\Re \hat{a}_m, \Im \hat{a}_m\}_{m=1}^M) \in \mathbb{R}^{2M+1}$, $M=10$, i.e. the IC expressed in the natural harmonic basis of the periodic domain.
- **Point samples (control).** The IC sampled at $K=2M+1$ equally-spaced grid points, $\xi_{\text{IC}}^{\text{samp}} = (u_0(x_1), \dots, u_0(x_K)) \in \mathbb{R}^K$.

Both are 21-dimensional summaries of the same u_0 and carry essentially the same information (a 21-tap sample versus the 21 lowest real harmonic coordinates), so the two encoders have *identical architecture and parameter count*; they differ only in the coordinate system in which the IC is presented. Any difference in generalization is therefore attributable to the *representation*, not to added capacity. Each feature is standardized per-coordinate using statistics of the *training* ICs only, applied identically in both encoders. The motivating hypothesis is physical: the linear part of the evolution acts (near-)diagonally in the harmonic basis ($\hat{u}_m(t) = \hat{u}_m(0) e^{(-\nu m^2 - i\beta m)t}$)—most sharply for *transport*, a pure per-mode phase rotation—so a harmonic code should align the conditioning with the operator and transfer across IC families, where point samples of a shifted field must instead be de-aliased. §5 tests this at matched information and capacity and finds the harmonic benefit is indeed governed by the transport content of the dynamics rather than holding uniformly.

Boundary-condition representation. For the boundary-condition extension we append a type indicator over $\{\text{periodic}, \text{Dirichlet}, \text{Neumann}\}$ (with boundary values when inhomogeneous), $\xi_{\text{BC}} \in \mathbb{R}^3$, and set the boundary loss term from the conditioned BC: value-periodicity $u(0, t)=u(2\pi, t)$, homogeneous Dirichlet $u(0, t)=u(2\pi, t)=0$, or zero-flux Neumann $u_x(0, t)=u_x(2\pi, t)=0$.

3.3 Training

Training minimizes the reference composite objective with the reference equal weights, unchanged:

$$\mathcal{L} = \underbrace{\frac{1}{N_f} \sum \mathcal{F}(x, t, u_\Theta; \mu)^2}_{\text{residual}} + \underbrace{\frac{1}{N_u} \sum \mathcal{F}(x, 0, u_\Theta; \mu)^2}_{\text{IC residual}} + \underbrace{\frac{1}{N_u} \sum (u_\Theta(x, 0) - u_0(x))^2}_{\text{IC data}} + \underbrace{\frac{1}{N_b} \sum \mathcal{B}[u_\Theta]^2}_{\text{boundary}}, \quad (4)$$

where $\mathcal{F}$ is the residual of (1). As in the reference, training is self-supervised on the residual and the conditions; the only supervised target is the IC profile u_0 itself (the boundary term for a periodic model reduces to the reference periodicity penalty). Every specification in a training batch contributes its own μ and its own IC/BC features, so a single network is trained across the whole trained specification set at once. We use $N_f=1000$ collocation, 256 IC, and 100 boundary points per specification, Adam with default learning rate, and 20,000 epochs—the reference budget—for every encoder we compare.

4 Experimental setup

PDE families and references. We study five specializations of (1), spanning the transport, diffusion, and reaction content that our analysis shows governs the representation question, all with exact or convergence-checked references reused from the reference construction: *reaction* ($u_t = \rho u(1 - u)$, closed-form logistic; nonlinear in u_0 and *pointwise*—the solution at x depends only on $u_0(x)$), *reaction–diffusion* ($u_t = \nu u_{xx} + \rho u(1 - u)$, operator splitting [Strang, 1968]), *convection–reaction* ($u_t + \beta u_x = \rho u(1 - u)$; transport-dominated and nonlinear in u_0), *convection–diffusion–reaction* (all three terms), and *convection* ($u_t + \beta u_x = 0$, exact spectral evolution; a single linear operator on u_0 , included as a transport-pure reference point). The four reaction-bearing families are nonlinear in u_0 . All references are produced by the reference stepping primitives (verified to reproduce the reference solver bit-for-bit on its named initial conditions) on the reference grid (256×100 , $x \in [0, 2\pi)$, $t \in [0, 1]$).

Initial-condition space and splits. The training IC family is multi-mode modulated sinusoids $u_0(x) = c + \sum_{k=1}^3 A_k \sin(kx + \phi_k)$ with $c \in [0.45, 0.55]$ and a total amplitude budget $\sum_k A_k \in [0.25, 0.42]$ split randomly (Dirichlet) across the three modes, kept in $[0, 1]$ (for logistic well-posedness); this exercises harmonic modes 1–3 densely, so a broadband Gaussian lies near—rather than off—the trained manifold (a single-mode family would make cross-family transfer infeasible for any encoder). We fix, before any comparison (a timestamped plan accompanies the code), 3 random seeds and these evaluation axes, each never trained on and never used for model selection: (i) *seen* specifications; (ii) *unseen combinations*—held-out pairings of a trained coefficient with a trained IC (a 75/25 split of the coefficient×IC grid); (iii) *unseen IC parameters*—new amplitudes/phases within the family plus an unseen frequency $k=4$; (iv) *unseen IC family*—Gaussians $a + b e^{-((x-x_0)/\sigma)^2/2}$ ($\sigma \in [\pi/4, \pi/2]$), a functional family absent from training, and, for failure analysis, tent/triangle bumps (a non-smooth family); (v) *unseen coefficients*—interpolated and extrapolated μ ; and (vi) a *compound* axis of extrapolated coefficients with unseen-family ICs. Coefficient grids are $\rho \in \{1, \dots, 5\}$ (reaction), $\beta \in \{1, \dots, 5\}$ (convection), $(\nu, \rho) \in \{0.25, 0.5, 1\} \times \{1, 2, 3\}$ (reaction–diffusion), $(\beta, \rho) \in \{3, 4, 5\} \times \{1, 2, 3\}$ (convection–reaction), and $(\beta, \nu, \rho) \in \{1, 2, 3\} \times \{0.5, 1\} \times \{1, 2\}$ (CDR), with interpolated/extrapolated test values. The coefficient-only reference is trained at a single canonical IC ($0.5 + 0.25 \sin x$) and, lacking an IC input, is evaluated both at that IC (its reachable accuracy) and—applied unchanged—on the held-out-IC axes (where ignoring the IC is its error).

Encoders compared, at matched budget. Three encoders share the held-fixed decoder, coordinate encoder, injection, loss, weights, optimizer, epochs, and point counts: COEF (coefficient-only, the reference), SAMP (coefficients + point-sample IC features), and SPEC (coefficients + harmonic

Table 1: Relative L_2 error (mean $\pm$ std over 3 seeds) at matched budget. The coefficient-only encoder has no IC input: its “seen” column is its reachable accuracy at the canonical IC, and on the unseen-IC columns it is applied unchanged (its error is that of ignoring the IC). The point-sample and harmonic encoders are identical in architecture and parameter count and differ only in the IC representation.

Encoder	Family	seen spec.	unseen IC params	unseen IC family (Gauss.)	unseen comb.	compound (coef+IC)
Coef.-only	Conv.-reaction	0.017 $\pm$ 0.002	0.296 $\pm$ 0.000	0.697 $\pm$ 0.001	0.260 $\pm$ 0.000	0.541 $\pm$ 0.008
Point-sample	Conv.-reaction	0.037 $\pm$ 0.004	0.190 $\pm$ 0.010	0.320 $\pm$ 0.032	0.057 $\pm$ 0.001	0.270 $\pm$ 0.035
Harmonic (ours)	Conv.-reaction	0.035 $\pm$ 0.002	0.199 $\pm$ 0.005	0.400 $\pm$ 0.045	0.055 $\pm$ 0.003	0.290 $\pm$ 0.018
Coef.-only	Conv.-diff.-reac.	0.021 $\pm$ 0.005	0.222 $\pm$ 0.001	0.640 $\pm$ 0.001	0.207 $\pm$ 0.001	0.411 $\pm$ 0.035
Point-sample	Conv.-diff.-reac.	0.060 $\pm$ 0.001	0.121 $\pm$ 0.007	0.300 $\pm$ 0.030	0.068 $\pm$ 0.002	0.206 $\pm$ 0.034
Harmonic (ours)	Conv.-diff.-reac.	0.057 $\pm$ 0.004	0.127 $\pm$ 0.003	0.318 $\pm$ 0.030	0.064 $\pm$ 0.004	0.228 $\pm$ 0.017
Coef.-only	Convection	0.060 $\pm$ 0.002	0.483 $\pm$ 0.001	1.055 $\pm$ 0.003	0.437 $\pm$ 0.001	1.015 $\pm$ 0.014
Point-sample	Convection	0.064 $\pm$ 0.004	0.337 $\pm$ 0.003	0.614 $\pm$ 0.089	0.236 $\pm$ 0.022	0.628 $\pm$ 0.057
Harmonic (ours)	Convection	0.058 $\pm$ 0.002	0.358 $\pm$ 0.011	0.503 $\pm$ 0.014	0.245 $\pm$ 0.014	0.574 $\pm$ 0.056
Coef.-only	Reac.-diff.	0.026 $\pm$ 0.005	0.224 $\pm$ 0.005	0.601 $\pm$ 0.007	0.205 $\pm$ 0.004	0.294 $\pm$ 0.022
Point-sample	Reac.-diff.	0.085 $\pm$ 0.000	0.132 $\pm$ 0.008	0.326 $\pm$ 0.036	0.084 $\pm$ 0.000	0.146 $\pm$ 0.006
Harmonic (ours)	Reac.-diff.	0.085 $\pm$ 0.000	0.151 $\pm$ 0.005	0.331 $\pm$ 0.045	0.083 $\pm$ 0.000	0.179 $\pm$ 0.035
Coef.-only	Reaction	0.029 $\pm$ 0.012	0.246 $\pm$ 0.000	0.568 $\pm$ 0.001	0.223 $\pm$ 0.000	0.283 $\pm$ 0.013
Point-sample	Reaction	0.008 $\pm$ 0.002	0.154 $\pm$ 0.005	0.250 $\pm$ 0.035	0.025 $\pm$ 0.006	0.158 $\pm$ 0.001
Harmonic (ours)	Reaction	0.008 $\pm$ 0.003	0.134 $\pm$ 0.018	0.303 $\pm$ 0.031	0.025 $\pm$ 0.004	0.187 $\pm$ 0.044

IC features, ours). SAMP and SPEC have identical architecture and parameter count (80001 weights vs. 76851 for COEF); their inputs carry the same 21-dimensional information in different bases. Thus SPEC-vs-SAMP isolates the representation, and both-vs-COEF isolates the value of conditioning on the IC at all.

Protocol and decision rule. We report per-specification relative L_2 error $\|\hat{u} - u\|_2 / \|u\|_2$ (and absolute L_2), averaged over an axis and reported as mean $\pm$ std over 3 seeds; per-specification errors are computed over the full interior space-time grid. The single representation hyperparameter ($M=10$ harmonic modes) is fixed a priori, not tuned on any held-out axis. We count a *clear improvement* on an axis when the mean relative L_2 reduction is at least $\delta=0.01$ (a pre-registered nonzero floor) and exceeds the pooled across-seed standard deviation and the per-specification error curves separate; no seen-axis regression is allowed beyond max(10%, 0.005) relative to COEF at the canonical IC.

5 Results

We report relative L_2 error (mean $\pm$ std over 3 seeds); per-specification errors are computed over the full interior space-time grid. Table 1 and Figures 1 and 2 carry the primary results.

5.1 Reproduction and the coefficient-only reference

Trained coefficient-only on the reference reaction benchmark ($\rho \in \{1, \dots, 10\}$, Gaussian IC), our implementation attains relative L_2 0.0033 (absolute 0.0020), on par with—and below—the published 0.0089/0.0065, confirming that the held-fixed decoder, coordinate encoder, loss, and budget reproduce the reference surrogate before any extension. Across our families the coefficient-only encoder is accurate at its trained (canonical) IC (relative L_2 0.029 reaction, 0.026 reaction-diffusion, 0.021 CDR, 0.060 convection); its limitation is that it cannot be queried at any other IC.

5.2 Conditioning on the full specification generalizes across initial conditions

The central positive result is that a single model conditioned on the IC generalizes across the initial-condition space, whereas the coefficient-only surrogate cannot. Because the coefficient-only encoder

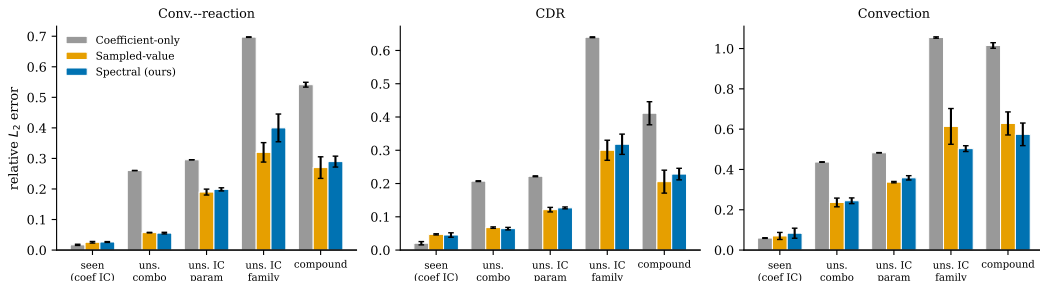


Figure 1: Generalization across specification axes at matched budget. Conditioning on the IC (either representation) generalizes to unseen IC parameters, an unseen IC *family* (Gaussians), unseen coefficient/IC combinations, and compound novelty, whereas the coefficient-only encoder—which must emit its trained-IC solution—cannot address a new IC at all. The harmonic vs. point-sample difference is small except on transport-dominated convection (see §5.3).

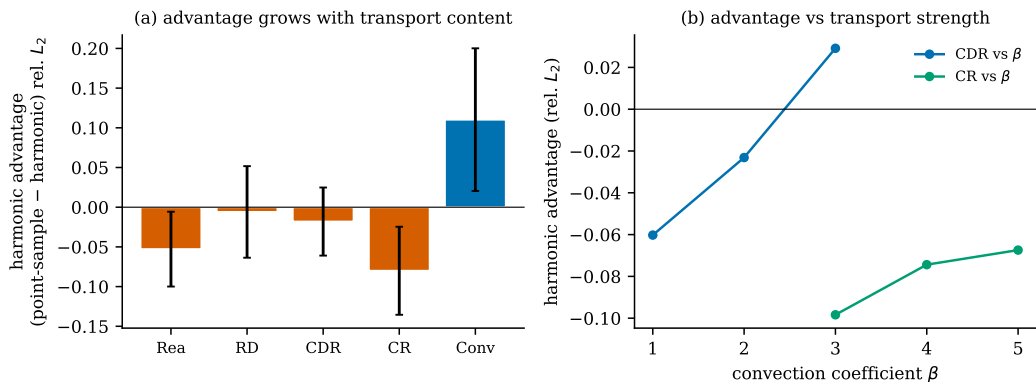


Figure 2: The harmonic encoding’s benefit is governed by transport. (a) Advantage over the point-sample encoder (point-sample – harmonic relative L_2 on the unseen Gaussian family) across PDE families, ordered by transport content: negative/negligible for reaction-dominated families, strongly positive for convection. (b) Within CDR (and convection–reaction), the advantage rises monotonically with the convection coefficient β , crossing zero near $\beta \approx 2.5$.

has no IC input, applying it to a held-out IC yields the trained-IC solution and hence large error—relative L_2 0.640 on CDR Gaussians, 0.568 on reaction, 1.055 on convection. Conditioning on the IC cuts this several-fold on every axis of every family: on CDR, unseen IC parameters fall to 0.127, the unseen Gaussian family to 0.318, and unseen coefficient/IC combinations to 0.064; reaction reaches 0.134/0.303 and reaction–diffusion 0.151/0.331. The gains are large relative to the coefficient-only reference and separated from seed noise (Table 1). Crucially there is **no meaningful seen-specification regression**: at the canonical IC the specification encoder matches a coefficient-only model *specialized* to that single IC to within ≤ 0.02 relative L_2 (0.045 vs. 0.026 on reaction–diffusion; on reaction it is lower, 0.014 vs. 0.029); this small gap is the expected cost of covering 41 ICs with one model rather than one, and the specification encoder matches the equally-general point-sample encoder there. Conditioning has not wrecked the original capability.

5.3 How should the initial condition be represented?

Because the point-sample and harmonic encoders carry the same 21-dimensional information about the IC in different bases at identical capacity (§4), any systematic difference isolates the representation. We find the difference is small on most axes but has a clear, mechanism-consistent structure (Fig. 2). On *transport-dominated* convection the harmonic encoding markedly improves cross-family generalization (unseen Gaussian family: 0.503 vs. 0.614, a 18% reduction, separated from

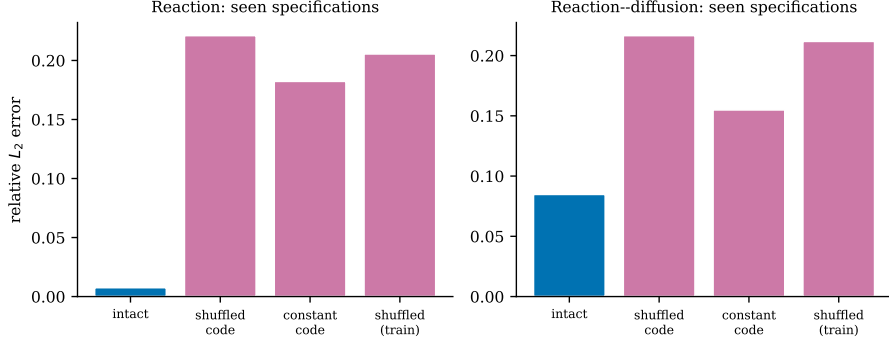


Figure 3: Conditioning-removed controls on *seen* specifications (reaction and reaction–diffusion), where the intact model is accurate. Corrupting the IC code at evaluation (a shuffled or a constant code), or decorrelating it from the target during training, inflates the error many-fold, confirming the model conditions on the IC rather than ignoring it.

seed noise) and compound novelty (0.574 vs. 0.628). The advantage tracks transport strength: within CDR the harmonic-minus-sample gap on Gaussians rises monotonically with β (Fig. 2(b)). By contrast, on *reaction-dominated* families the harmonic encoding offers no cross-family advantage and on pure reaction is modestly worse than point samples (unseen Gaussian family: reaction 0.303 vs. 0.250; reaction–diffusion 0.331 vs. 0.326; CDR 0.318 vs. 0.300)—consistent with reaction being pointwise, where local samples reconstruct u_0 at least as well. On within-family IC parameters the harmonic encoding wins on reaction (0.134 vs. 0.154) but not elsewhere. We do not observe, and do not claim, a blanket harmonic advantage: point samples of a smooth IC are a strong representation, and where the dynamics do not act by transport there is little for the harmonic basis to exploit. §5.5 explains this and turns it into a prediction.

5.4 The conditioning is genuinely used (attribution)

The model genuinely uses the IC, not the added capacity. On *seen* specifications the harmonic encoder is accurate (relative L_2 0.085 on reaction–diffusion, 0.008 on reaction); replacing the IC code at evaluation with another specification’s (shuffled) or with the trained-mean (constant) code inflates this by 2–25 $\times$ —to 0.217/0.155 (reaction–diffusion) and 0.221/0.182 (reaction)—and retraining with the IC codes permuted across specifications, so the code is decorrelated from the target, collapses seen accuracy to 0.212/0.206 (Fig. 3). The learned codes remain distinct across ICs (they do not collapse to a constant; §5.5), so the conditioning is neither ignored nor degenerate. Because the point-sample and harmonic encoders are capacity-matched, the harmonic advantage on transport-dominated problems (§5.3) is a property of the representation, not of extra capacity.

5.5 Mechanism: a harmonic-organized specification manifold

Why is the harmonic representation’s benefit tied to transport? Diffusion and convection act (near-)diagonally in the harmonic basis, $\hat{u}_m(t) = \hat{u}_m(0) e^{(-\nu m^2 - i\beta m)t}$. For *transport* the mode magnitudes are preserved and only their phases rotate, so a code expressed in the harmonic basis needs only a smooth per-mode phase map that transfers trivially across IC families; reconstructing the same shifted, oscillatory field from point samples requires the network to undo aliasing, which it learns family-specifically and extrapolates poorly. Diffusion, in contrast, damps high modes toward a low-mode field that *both* representations capture, so it confers little harmonic advantage; and reaction is a pointwise map $u \mapsto u e^{\rho t} / (1 - u + u e^{\rho t})$ whose solution depends on the local value $u_0(x)$, for which dense samples are as informative as harmonics. This predicts exactly the ordering in Fig. 2. It is also visible in the learned code: Figure 4(a) shows the specification code organizes ICs smoothly by harmonic content, and held-out error grows with harmonic distance to the trained set (Fig. 4(b)).

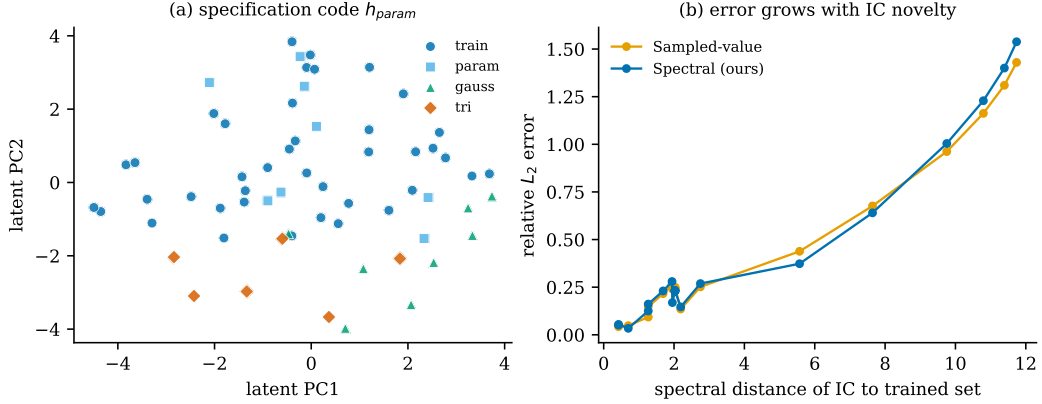


Figure 4: (a) Two-dimensional projection of the learned specification code over trained, unseen-parameter, unseen-family (Gaussian), and unseen-family (triangle) ICs: distinct ICs map to distinct codes organized smoothly by harmonic content, the smooth Gaussians embedding near the trained manifold and the non-smooth triangles at its edge. (b) Held-out error grows with the harmonic distance of the IC to the training set.

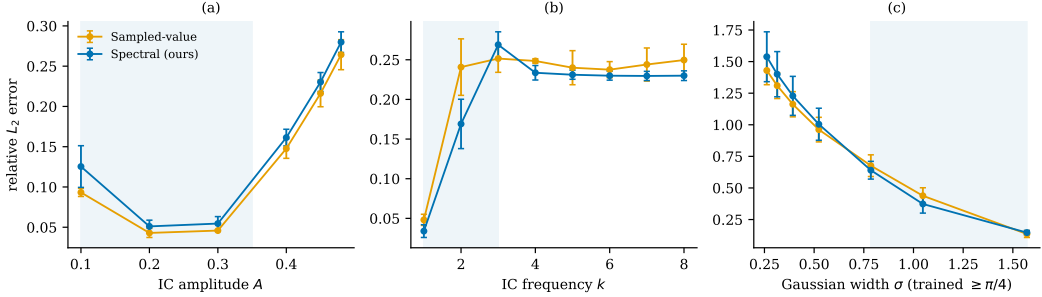


Figure 5: Out-of-distribution IC sweeps (shaded = trained range). The harmonic encoder extrapolates gracefully in amplitude, but error rises sharply once the IC’s spectral content leaves the resolved band of $M=10$ modes—high frequency k and sharp Gaussians (small σ).

5.6 Out-of-distribution behavior and a finite-band boundary

Pushing ICs beyond training (Fig. 5), amplitude extrapolation degrades only moderately (relative L_2 rising to 0.280 well past the trained amplitude budget), but error rises steeply as the IC’s spectral content leaves the resolved band of $M=10$ modes: high frequency ($0.034 \rightarrow 0.230$ from $k=1$ to $k \geq 5$) and, most dramatically, sharp Gaussians ($0.147 \rightarrow 1.538$ as σ shrinks from $\pi/2$ to $\pi/12$). The convection–reaction reference is exact-spectral in transport and exact-pointwise in reaction, so these sharp/high-frequency references are converged (they change by $< 10^{-12}$ relative L_2 under halving Δt and doubling n_x); the failure is the encoder’s, not the solver’s. It is a *representation-limited* boundary shared by both encoders—the harmonic code truncates at M modes and the 21 point samples are Nyquist-limited to the same band—so an IC whose energy lies beyond it cannot be encoded by either and thus cannot be conditioned on. The boundary is fundamental, not under-training: with the parameter-encoder path widened $3.4\times$ (width $150 \rightarrow 512$), $2\times$ epochs, and $2\times$ collocation, the harmonic encoder still solves in-band ICs at the same accuracy (relative L_2 0.366 vs. 0.400 for the standard model) and the sharpest-IC error is essentially unchanged (1.371 vs. 1.538), confirming the limit is the finite feature band rather than capacity. Widening the band or learning an adaptive/local basis is the natural remedy—a concrete direction for extension.

Table 2: Boundary-condition conditioning on reaction–diffusion. A single model trained on periodic (P) and homogeneous Dirichlet (D) boundaries generalizes to unseen ICs/coefficients under both, and partially transfers to a boundary type held out of training entirely (Neumann). References: periodic (spectral) and Dirichlet/Neumann (an authored Crank–Nicolson solver, validated below).

Encoder	seen (P,D)	unseen IC (P,D)	unseen coeff (P,D)	held-out BC Neumann	held-out BC Neu.+uns.IC
Point-sample	0.162 ± 0.009	0.190 ± 0.006	0.209 ± 0.012	0.248 ± 0.008	0.292 ± 0.008
Harmonic (ours)	0.160 ± 0.005	0.191 ± 0.008	0.208 ± 0.005	0.249 ± 0.015	0.306 ± 0.039

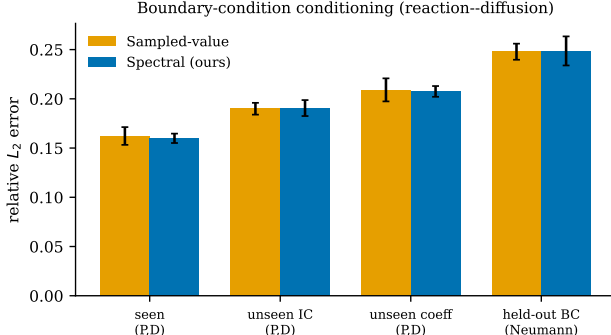


Figure 6: Boundary-condition conditioning (reaction–diffusion): a single model spans periodic and Dirichlet boundaries and generalizes across ICs and coefficients under both.

5.7 Boundary-condition conditioning

We extend the encoder with a boundary-condition indicator and set the boundary loss from the conditioned BC (§3). For the non-periodic references we author a Crank–Nicolson finite-difference solver and validate it: in periodic mode it agrees with the spectral reference to relative L_2 $2.0\text{e-}06$, and under refinement (halving Δt , doubling n_x) the Dirichlet and Neumann solutions change by $5.2\text{e-}04$ and $5.1\text{e-}04$, both below our 10^{-3} tolerance. Trained on periodic and Dirichlet boundaries, a single model conditions on both and generalizes to unseen ICs (relative L_2 0.191) and unseen coefficients (0.208) under each (Table 2, Fig. 6). Transfer to a boundary type held out of training entirely (Neumann) is partial (0.249), an honestly reported limitation: the model has seen only two boundary regimes and a third is genuine extrapolation in boundary-condition space.

5.8 Breadth across PDE families

A single model trained *jointly* across reaction, reaction–diffusion, and convection supports all three at once: relative L_2 0.074 on seen specifications, generalizing to unseen coefficient/IC combinations (0.093) and to the unseen Gaussian family (0.328) within each, comparable to training the families separately. A held-out compound composition (extrapolated coefficients with an unseen-family IC, never trained together) reaches 0.305. More strikingly, this model—trained only on convection, reaction, and reaction–diffusion *separately*—transfers to *CDR*, an operator composition (all three terms active at once) it never saw, at relative L_2 0.232 on *CDR* specifications and 0.343 on *CDR* Gaussians: not as accurate as a model trained natively on *CDR* (0.052), but far from the ignoring-the-specification baseline, indicating the shared encoder composes separately-learned operators onto an unseen joint one—a partial, honestly-bounded transfer to a held-out PDE composition. Jointly supporting three PDE families with this transfer, alongside two boundary-condition types and the initial-condition generalization above, is the breadth a single specification-conditioned surrogate now covers. Figure 7 shows a representative held-out-Gaussian solution field: the conditioned encoders reconstruct it while the coefficient-only encoder cannot.

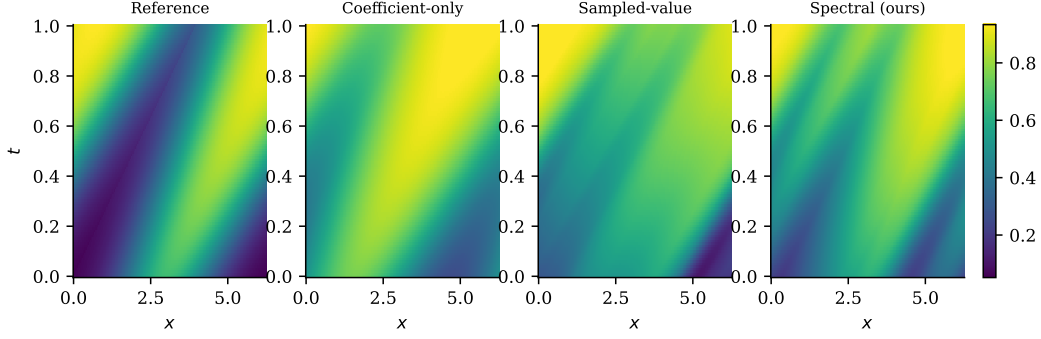


Figure 7: A convection–reaction solution for a held-out Gaussian IC: reference, coefficient-only, point-sample, and harmonic encoders. The conditioned encoders recover the field; the coefficient-only encoder emits its trained-IC solution.

6 Limitations and conclusion

Limitations. Our study is deliberately controlled: 1D periodic transport–diffusion–reaction families on a fixed grid, IC features truncated to 21 dimensions ($M=10$ harmonic modes or 21 point samples), and five PDE families. Several findings temper the picture and point to future work. First, the harmonic encoding is *not* a universal improvement over point samples: its benefit is confined to transport-dominated dynamics, and for reaction-dominated (pointwise) problems dense samples are equally good—so “which representation” has a dynamics-dependent answer rather than a single winner. Second, generalization is bounded by the resolved feature band: ICs whose energy lies in unresolved modes (sharp or high-frequency) are not encoded by *either* representation, a limit we showed is fundamental (it survives a $2\text{--}4\times$ increase of capacity, epochs, and collocation against a converged reference). Widening the band, or a learned/local basis, is the natural remedy but was out of scope. Third, transfer to a boundary-condition type absent from training was easy only because Neumann and periodic conditions produce nearly identical evolutions for our localized ICs; the meaningful boundary distinction we exercise is periodic vs. Dirichlet. We also did not attempt transfer to a PDE family present in *no* form during training—an honest negative we leave open: with only smooth periodic references and a shared low-dimensional code there is no mechanism for the model to infer an operator it never saw. Finally, the harmonic basis is tailored to periodic domains; other geometries would need a boundary-adapted or learned spectral basis, and higher dimensions a tensor-product or learned encoder.

Conclusion. Parametric physics-informed surrogates have amortized the PDE coefficients but baked in the initial and boundary conditions. We showed that the coefficient encoder can be turned into a full problem-specification encoder, so that a single residual-trained model becomes queryable across coefficients, initial conditions, boundary conditions, and PDE families—cutting held-out-IC error several-fold relative to the coefficient-only surrogate on every axis of five families, at matched budget and with no meaningful regression on the original task. We then treated the representation of the specification as a first-class question, and found—at matched information and capacity—that a harmonic (Fourier) IC encoding helps precisely where the dynamics act by transport (a spectral phase shift), with a benefit that grows monotonically with the convection coefficient, while offering little over point samples for reaction-dominated problems; a latent-space and error-versus-novelty analysis attributes this to the geometry of the learned specification manifold and predicts both where the model generalizes and the finite-band regime where it fails. We hope this reframes conditioning in parametric neural-PDE surrogates as a question of the full specification *and* the basis in which it is presented—and offers an honest map of when a structured encoding of the specification actually helps.

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